

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam. No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer the questions in the space provided. All questions carry equal marks. Max Marks: 10

1. Arrange these correctly when the system boots: BIOS, Power On, Boot Loader, OS image.

Power On → BIOS → Boot Loader → OS Image

2. Which of these is faster to boot - init-based (System V) or Systemd-based Linux systems?

Systemd-based

3. Which Linux directory contains per-process information such as virtual memory, stack, limits, etc?

/proc

4. Type the command to get the process ID of the current shell.

echo \$\$

5. Within Vi editor, mention the command to replace all occurrences of Average with Good

:%.s/Average/Good/g

6. How do you locate multiple binaries of the same name in Linux using shell/terminal?

locate canary

7. How do you help Bhargav list all shells present in the system?

cat /etc/shells

8. Aman is struggling to recall a command. However, he remembers it starts with ls. Help him to locate such commands.

compgen -A command ls

9. On Ubuntu, Sayuj cannot recall if set the root password. How do you confirm if no root password is set?

sudo -i ⇒ passwd -s ⇒ check np in the output
OR sudo passwd -s

10. Piyush has a dual-booted system with Windows and Linux (Ubuntu). His system does not show the correct time as per the timezone set. He has booted into Ubuntu. What command should he use to synchronize the clock?

timedatectl

set-ntp true

timedatectl

set-local-rtc 1 --adjust-system-clock

11. How do you find file systems available on the Linux systems? Multiple file systems can co-exist.

sudo fdisk -l OR cat /etc/fstab OR df -hT

12. Help Himanshu add a cron entry to sync attendance data from devices to the server at 11 pm every day.

The user is heman and script is located at /home/heman/syncatt.sh

00 23 * * * heman { cd && ./syncatt.sh }

13. Name two package managers used in Linux systems, along with the distribution name.

apt, dpkg ⇒ Ubuntu / Debian OR rpm, yum ⇒ RHEL, Fedora
synaptic

14. How can Puru delete the current and the following 10 lines in the vi editor?

:.,+10 d OR 11 dd

15. Sahil remembers the command name. However, he is unsure about some of the command's options and arguments. Give him two options to find what he needs on the Linux system.

man, help, info pages

16. Anubhav wants to see which process takes the maximum CPU time and memory at any instant in Linux. Suggest one option.

top

17. Parth discovered that his disk partition was almost full. He recalls one large file in his home directory that he can delete, however rm is not working. Help him to get rid of the contents of /home/parth/game.tar

>/home/parth/game.tar

18. Shubham wants to share some files on the same network using an HTTP server. He has Python installed. Help him.

python3 -m http.server

19. How do you find out the space APT Cache is taking in human-readable size? Hint. /var/cache/apt

du -hs /var/cache/apt

20. How do you list the partitions on the disk? Command please.

sudo fdisk -l

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1. How can you **search** a log file for ERROR and print five lines before and five lines after the matching line?

grep -C 5 "ERROR" logfile

2. How will you **trace** nameservers from root nameservers to the nearest authoritative DNS?

dig +trace auth-dns-server-or-domain

3. Name/mention one **public DNS** IP address. Hint: all four digits are the same

8.8.8.8 or 1.1.1.1

4. Name/mention one **private IP** address **range**. Hint: campus network

10.0.0.0 - 10.255.255.255, 172.16.0.0 - 172.31.255.255, 192.168.0.0 - 192.168.255.255

5. Name one thing that **lscpu** shows that **/proc/cpuinfo** does not show.

cache stats, vendor details

6. Name one command with an option to avoid accidental **file overwrite**. Hint: change attributes

chattr +i file OR chattr +a file (append-only)
(immutable)

7. Name the command with options to find out the **top memory-hungry processes**. Hint: instantaneous top

top -bn1 -o %MEM | head -n 15

8. Name one command to generate an **audit summary** of system logs. Hint: AuditD

sudo aureport

9. Write a **tcpdump** command to capture DNS traffic (port 53) and write to dns_capture.pcap file.

sudo tcpdump port 53 -w dns_capture.pcap

10. Write the **Network Mapper** command to scan the first 1024 ports of the site myexample.com

nmap -p 1-1024 myexample.com

Indian Institute of Technology Jammu
Linux System Administration and Virtualization (ID-2-04(MO))
Mid Semester Exam (Session 2025-26)

Student ID: _____ Student Name: _____

Instructions: Usage of unfair means (UFM), including Mobile and Internet, is not allowed in the exam.

No cheating. Any kind of cheating will get you a straight a Fail grade in the course.

Answer the questions in the space provided. Q1-10 carry 1 Mark each, Q11-40 2 Marks and Q41-44 carry 10 Marks each.

1. Which one of the following is not a shell?

- (A) /bin/bash (B) /bin/dash
(C) /bin/rbash (D) /bin/rdash

[D]

2. On /bin/bash, what does the command **!!** do?

- (A) List history of commands (B) Repeat the last command
(C) Print the shell process ID (D) Clear the history of commands

[B]

3. Which of the following command locates the binary, source, and manual page files for a command?

- (A) plocate (B) whereis
(C) which (D) whatis

[B]

4. What is the numeric value of permissions for this file?

crw-rw-rw- 1 root root 1, 3 Feb 20 17:24 /dev/null

- (A) 666 (B) 766
(C) 777 (D) 444

[A]

5. A log file keeps updating and growing. How can we view its content while it is still getting updated?

- (A) head file (B) tail -f file
(C) tail -z file (D) cat -t file

[B]

6. Which of the following is incorrect about bash builtin cd?

- (A) cd takes to /home directory (B) cd ~ takes to home directory
(C) cd .. takes to parent directory (D) cd - takes to previous directory

[A]

7. How to get a list of all signals on the Linux, name, and number?

- (A) kill -l (B) kill -a
(C) kill -9 -a (D) kill %1

[A]

8. Which one of the following is an LTS release of Ubuntu?

- (A) 22.10 (B) 23.04
(C) 23.10 (D) 24.04

[D]

9. In the ls -lrt output -rwsr-xr-x, what is **s** meant for?

- (A) symbolic link (B) sticky bit
(C) shared object (D) shell script

[B]

10. Execute the last command in Ubuntu terminal that began with com

- (A) runlast com (B) !com
(C) ^com (D) :&com

[B]

11. On Ubuntu, rps cannot recall if he set the root password. How do you confirm if no root password is set?

`sudo -i`

`passwd -S` $\Leftarrow$ check output, NP - no password set

12. rps has a dual-booted system with Windows and Linux (Ubuntu). His system does not show the correct time as per the timezone set. He has booted into Ubuntu. What command should he use to synchronize the clock across OSs? Mention complete command.

`timedatectl set-local-rtc 1 --adjust-system-clock`

`timedatectl set-ntp true`

13. cat command can be used to merge two files, file1 and file2. Write the command.

`cat file1 >> file2`

14. rps has a compressed file ~/Downloads/json.zip and he wants to know its compressed size, uncompressed size, compression ratio, and name of uncompressed file. Help him with the correct command.

`zcat -l ~/Downloads/json.zip`

`gzip -l`

15. rps wants to view line numbers 1260 to 1265 in the file filename.txt. Help him with the correct command.

Hint: He recalls something from the head and tail.

`head -1265 filename.txt | tail +1260` ✓

or `tail +1260 filename.txt | head -6`



16. rps wants to share some files on the same network using an HTTP server. He has Python installed. Help him with the correct command to start the server.

`python3 -m http.server`

17. Using the read command, how can we show a prompt to "Enter Name" and read the name into a variable sname?

`$ READ_COMMAND_HERE` $\Leftarrow$

Enter Name: Roger Southwood

`$ echo $sname`

Roger Southwood

`read -p "Enter Name:" sname`

18. Complete if statement in this bash script to match the output from echo.

`file1=$1`

`file2=$2`

`IF_STATEMENT_HERE` $\Leftarrow$

`echo "$file1" is newer than "$file2"`

`fi`

`if [[ $file1 -nt, $file2 ]]; then`

19. How can standard output and standard error be directed to the file log.txt?

&> log.txt

20. How to output the command output of lsb_release -a to the terminal and a file called release.txt also?

lsb_release -a | tee release.txt

21. On /bin/bash terminal, what is the output of \$ yes

continuous text y is printed until interrupted, liveness

22. What is the difference between these two ways of executing a bash script?

\$ bash -v script.sh

\$ bash -x script.sh

bash -v display command before running then execute and output

bash -x display command after processing, it expands variables

23. How will you count the number of processors on the Linux system using cpuinfo? Command, please.

cat /proc/cpuinfo | grep -i "processor" | wc -l

24. How can we see the process memory layout of process ID 54321 using the proc interface?

cat /proc/54321/maps

25. Inside vi editor, in command mode, how will you remove all occurrences of control characters ^V

:%s/^V//g

26. How can we list out all the processes that belong to the logged-in user?

ps -aef | grep \$USER

27. For the local/current session in the terminal, how can we modify the PATH variable to search the home directory first for any executable?

export PATH=\$HOME:\$PATH

28. How can we find out the Audio Device details on the Linux system? Hint: the command starts with ls

lspci -v | grep -AT -i "audio"

29. How will you grant read permission to the group users and other users?

chmod go+r <file>

30. How can we find out multiple users that belong to a group called adm using command.

cat /etc/group | grep -i "adm"

31. Write down permissions in numeric representation from the partial output shown.

lrwxrwxr-x 775 drwxr-xr-x 755
 -r----- 400 -rw-r--r-- 644
 prw----- 600 srw-rw-rw- 666

32. Everything is file in Linux. The output of ls -lrt, column 1 lists file type. Fill in the blanks:

Regular file: - Directory: d
 Symbolic links: l Character device: c
 Block device: b Socket: s

33. A website URL <https://www.mysite.com/uploads/> shows all files uploaded by users. Anyone can delete them using HTTP DELETE [mysite.com/uploads/file](https://www.mysite.com/uploads/file) URLs. What is the issue here with the uploads directory, and how to fix it?

Issue uploads directory has write permissions set for others

Fix revoke write permissions for others

34. How can we query all executables that have the sticky bit set? Hint: find

find / -perm /1000 1000 SUID 4000 2000 1000
1000 ← sticky bit SUID SGID Sticky

35. The user has actually forgotten the command. He recalls it starts with ip. Show him all commands that start with ip so that he may recall or try them. Mention complete command with options.

compgen -A command ip OR compgen -c ip

36. How will you schedule a bash script to sync attendance /home/rps/sync_attendance.sh every night at 11 pm, except Sunday?

00 23 * * 1,2,3,4,5,6 rps {cd && ./sync_attendance.sh}

37. Mention two commands to list out both file systems and mount points.

38. How can rps create 100 files in /tmp directory using touch? File names should be named as app001.txt, app002.txt, app003.txt,.....,app099.txt, app100.txt.

touch /tmp/app{001..100}.txt

39. Why is a non-root user able to execute a command owned by root?

40. Why do system administrators prefer "su -" over a simple "su" command to substitute a user?

41. Write a bash script to calculate sum of natural numbers from 1 to n and n is command line argument. If n is not supplied, return error code 255. If n is not a non-negative number, return error code 254.

42. Write a bash script to check format of an input text as a three word name using regex. Each word begins with a capital letter and there is one space between words. There is no trailing space. Valid names are:

"S Arun Kumar", "Sam Arun Kumar", "S Arun", "Sam Arun K". "S A K",

While "SArunKumar", "S arun kumar", "arun kumar" are not in acceptable format.

~~#!/usr/bin~~

#!/usr/bin/bash

name=\$1

regexName="^[A-Z][a-z]* (\s[A-Z][a-z]*){0,2}\$"

if [[\$name =~ \$regexName]]; then

echo -e "Name format is valid"

else

echo -e "Name format is not valid"

fi

43. Write a bash script to check the format of an input as a valid date in DD-MM-YYYY format. Use regular expression.

```
#!/usr/bin/bash
```

```
date=$1
```

```
regexDate="^[0-9]{2}-[0-9]{2}-[0-9]{4}$"
```

```
if [[ $date =~ $regexDate ]]; then
```

```
    echo -e "date is valid"
```

```
else
```

```
    echo -e "date format is not valid"
```

```
fi
```

additionally

```
day=${BASH_REMATCH[1]}
```

```
month=${BASH_REMATCH[2]}
```

```
year=${BASH_REMATCH[3]}
```

```
echo -e "Day: $day, Month: $month, Year: $year"
```

44. Write a bash script to check the format of an input as valid IP address in w.x.y.z format. Each of w,x,y,z can be 1, 2 or 3 digit. Use regular expression.

```
#!/usr/bin/bash
```

```
ip=$1
```

```
regex_pattern="^([0-9]{1,3})\.([0-9]{1,3})\.([0-9]{1,3})\.([0-9]{1,3})$"
```

```
if [[ $ip =~ $regex_pattern ]]; then
```

```
    echo -e "IP format is valid"
```

```
else
```

```
    echo -e "IP format is not valid"
```

```
fi
```

alternate pattern

```
"^([0-255])\.([0-255])\.([0-255])\.([0-255])$"
```

Indian Institute of Technology Jammu
Linux System Administration and Virtualization (ID-2-04(MO))
End Semester Exam (Session 2025-26)

Student ID: 20 Student Name: _____

Instructions: The use of unfair means (UFM), including Mobile Devices and the Internet, is not allowed during the exam. No cheating. Any kind of cheating will get you a straight Fail grade in the course.

Answer all the questions in the space provided. Q1-15 are of 1 mark each. Q16-25 are of 2 marks each.
Max Marks: 35 Max Duration : 3 hours (180 minutes).

1. Event logs of the Linux system are stored at:
A) /var/log B) /etc/gub.conf C) /proc/stat D) /usr/sbin [**A**]
2. The option to run an interactive command on the remote system without a session disruption is:
A) nohup B) Run without stack C) Run without /proc D) ssh [**A**]
3. Which file system can be used to change certain kernel parameters at runtime using the sysctl command?
A) procfs B) sysfs C) btrfs D) ext4 [**A**]
4. The file system for the Linux root partition (/) is:
A) ext4 B) squashfs C) isofs D) tmpfs [**A**]
5. What is the utility of retrieving possible options for autocompletion of commands, builtins, arguments, etc.?
A) compgen B) complete C) which D) locate [**A**]
6. What is the primary role of the "inode" in a UNIX-based file system like EXT4?
A) It stores file metadata, such as file permissions, ownership, and pointers to data blocks
B) It stores file data directly
C) It tracks deleted files in the system
D) It acts as the root directory for the file system [**A**]
7. What is the main function of the /etc/shadow file in UNIX/Linux?
A) It stores user passwords in an encrypted format
B) It stores information about the user's home directories
C) It contains system logs and crash dumps
D) It tracks the system's performance metrics [**A**]
8. Which partitioning scheme uses a GUID Partition Table **GPT** instead of an MBR?
A) Windows FAT16, FAT32 based partitions
B) Modern UEFI-based systems
C) Linux ext2, ext3 based partitions
D) Legacy BIOS systems [**B**]
9. Which is a characteristic of solid-state drives (SSDs) compared to traditional hard drives (HDDs)?
A) Slower read/write speeds
B) Less reliable storage
C) Uses spinning disks
D) No moving parts [**D**]

10. Analysis of Linux shell history (.bash_history) shows a command **rm important_file.txt**. However, file recovery tools are unable to find any trace of this file. What is the most plausible explanation?
- A) The rm command on Linux always securely wipes data.
 - B) The file system did not support journaling.
 - C) The .bash_history file cannot be trusted.
 - D) The file was renamed before deletion.

[C]

11. While listing hidden files, ls -al command also lists the current directory (.) and the parent directory (..). How can we prevent the display of these two directories in the output when handling hidden files using the 'ls'? Say the name of directory is /home/rps/docs/

ls -Al /home/rps/docs

12. What will this command show? cat /proc/self/maps

Process memory layout of cat command executing this.
/usr/bin/cat code, text part, heap, shared libs, stack, etc.

13. Bash history:

- a. How can we prevent commands from being appended to history?

Add one or more spaces before the command.

eg. \$ __ cat /etc/shadow

- b. How to search bash history for a command containing "bhoot"?

Type ctrl+r (recursive reverse search) and type bhoot

14. Using sed, how can we add a prefix string "2025" at the beginning of each line in a file called paper.txt?

sed -i 's/^/2025/' paper.txt

15. You are given a file id.txt that contains students' IDs. Using sed, how can you add a suffix string "@iitjammu.ac.in" at the end of each line in this file?

sed -i 's/\$/@iitjammu.ac.in/' id.txt

16. Using secure copy, mention the command to

- a. Transfer a file, handout9.txt to a remote server 10.10.180.118 in the home directory /home/rps of user rps

scp handout9.txt rps@10.10.180.118:/home/rps/

- b. Fetch the file secret9.txt from the remote server 10.10.190.128, which resides in the home directory /home/guru of the user guru. The command should download this file to your home directory with the name as endsem.txt

scp guru@10.10.190.128:/home/guru/secret9.txt ~/endsem.txt

17. root user:

- a. How to check if the root user password is set on Linux?

\$ sudo -i and on next prompt type passwd -S

root@# passwd -S check output

root NP ← no pass is set
P ← passwd is set
L ← passwd is locked

- b. You forgot the root password. How to delete it once you find out it was set?

on root prompt type passwd -d root to delete password.

18. Cron jobs:

What do these two lines in /etc/crontab do?

- a. 0,10,20,30,40,50 * * * * rps { cd /home/rps/lav && touch a.out }

Every 10th minute, timestamp of file a.out is updated
in /home/rps/lav directory. Initially, once it is created.

- b. 00 7,13,19 * * * root { cd /home/rps/backup && ./run_backup.sh }

Every 7am, 1pm, 7pm script run_backup is executed
from /home/rps/backup directory

19. Write two commands each for file attributes:

Display the file attribute of **endsem.pdf** file

(1) lsattr endsem.pdf

(2) pdftinfo endsem.pdf

Display the file attribute of **answers.png** file

(1) lsattr answers.png

(2) identify answers.png

20. On a system, Linux/Ubuntu is installed.

- a. How to open these files using the same COMMAND? We do not know what the default application user has installed.

```
$ COMMAND beamer.pdf
$ COMMAND calvinEducation.jpeg
$ COMMAND thankyou.png
$ COMMAND latex-package.zip
$ COMMAND https://saral.iitjammu.ac.in
```

*+ replace COMMAND with
xdg-open*

- b. How to send a desktop notification "This is LSAV endsem exam."

notify-send "This is LSAV endsem exam."

21. SSL/TLS Certificate:

- a. While setting up an SSL/TLS certificate for the Election Commission server ec.iitjammu.ac.in, which two files are required on the server?

Certificate file .crt file

Private key file .key file

- b. How will you check the TLS version enabled and the SSL/TLS certificate expiry?

ssllscan ec.iitjammu.ac.in

*↳ check ssl version(s) enabled and validity date
at the end of this output.*

22. Attempt - File deletion and modification:

- a. The system has accumulated journals and should be cleaned; journalctl shows 10 GB of used space.

```
$ journalctl --disk-usage
```

Archived and active journals take up 10.0 GB in the file system.

How will you free up this space?

sudo journalctl --vacuum-time=1d

- b. Due to editing multiple files without using version control, you cannot determine how many files were changed recently. How will you determine which files have been modified within the last 24 hours? Command please.

`sudo find / -mtime 1` OR

`sudo find / -mmin -1440`

OR

Attempt - Vi Editor:

a) How to remove ^M (a single control character) in a file at the end of each line?

in command mode use `:%s/^M//g`

for this press `ctrl+v` `ctrl+m`

b) How to copy a block of text and paste it elsewhere in the vi editor?

23. Attempt - Curl:

a. What will be the output of this command?

`curl -v -o /dev/null -s -w '%{time_total}s %{num_connects} \n' https://sarat.iitjammu.ac.in/`

Total time for request and response for curl and the number of connections made.

b. How will you check HTTP response headers for the same URL using curl?

`curl -v https://sarat.iitjammu.ac.in`

OR

Attempt - Convert and Merge:

c. How to convert an image hate.png to PDF love.pdf

`convert hate.png love.pdf`

d. How to merge three PDF files file1.pdf, file2.pdf and file3.pdf into one pdf file merged_file.pdf

`pdftk file1.pdf file2.pdf file3.pdf cat output merged-file.pdf`

24. A directory `/home/rps/Downloads/` contains several WhatsApp Image files with jpeg extension, e.g.

'WhatsApp Image 2023-10-17 at 9.38.26 AM.jpeg'

'WhatsApp Image 2024-04-17 at 11.03.43 PM.jpeg'

'WhatsApp Image 2024-11-07 at 4.00.01 PM.jpeg'

'WhatsApp Image 2025-01-11 at 12.37.26 PM.jpeg'and so on.....

Write a script to rename them to `image-001.jpeg`, `image-002.jpeg`, `image-003.jpeg` and so on in a new directory `/home/rps/images`

25. Write a bash script to create 10 users, namely user1, user2, user3,...,user10, set their password as p@\$\$w0rd, and this will be shared with them. Now, set the password to expire so that users are prompted to change their password upon login. Script should output users created, and the password set to change upon login.



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You are required to show output /execution for each task. Max Marks-15

1. Install VirtualBox and download a Linux distribution (Ubuntu/CentOS/Kali). Create a VM with at least 2 GB of RAM, and allocate suitable resources to run this Linux distribution. After installation, and login, show the following:
 - a. Number of processors in it - using commands only
 - b. Memory installed and available - using commands only
 - c. File system - names and partitions - using commands only
 - d. List out all files in this VM with set user ID bit set - using commands only
 - e. Transfer files from and to this VM and your local desktop/laptop system

(5 Marks)
2. Backup and schedule job:
 - a. Write a shell script to create a backup directory with date, e.g. /backup/YYYY-MM-DD
 - b. Copy /var/log and /etc directory into it using this script
 - c. Create a tar.gz file, e.g. YYYY-MM-DD.tar.gz and then remove backup directory that was created in the same script
 - d. Schedule a backup using cron job to run at 4 pm everyday

(5 Marks)
3. Show the highest TLS version enabled on saral.iitjammu.ac.in and when the SSL certificate is set to expire.

(1 Marks)
4. How do you find files modified on the system/V in the last two hours? Hint: 120 minutes.

(1 Marks)
5. Show all sub-domains of iitjammu.ac.in

(1 marks)
6. Search "ERROR" or "Critical" in log file /var/log/syslog using regex with grep and show 5 lines before and after the occurrence.

(2 Marks)

